

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 3 Winter 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF10THOEM05E - INTERNET OF THINGS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12**
- 1 Which of the following best defines Cyber-Physical Systems (CPS)?
A. A software-only system used for documentation.
B. A system that integrates computation, networking, and physical processes
C. A social network for industrial workers. D. A cloud storage system for files
 - 2 Which of the following is a common short-range wireless technology used in IoT for low-power communication?
A. Ethernet. B. Bluetooth Low Energy (BLE). C. HDMI. D. SATA
 - 3 Which of the following is the primary role of an IoT gateway?
A. To replace sensors completely
B. To connect IoT devices to the cloud/network and perform protocol translation
C. To generate electrical power for IoT nodes.
D. To design user interfaces for mobile apps
 - 4 Why is edge computing important in many IoT applications?
A. It increases network congestion
B. It processes data closer to the device, reducing latency and bandwidth use
C. It eliminates the need for sensors.
D. It works only when the internet is unavailable
 - 5 Which information system is mainly designed to support top-level management by providing summarized reports and long-term trends?
A. Decision Support System (DSS). B. Transaction Processing System (TPS).
C. Executive Support System (ESS). D. Manufacturing Execution System (MES)
 - 6 A factory uses a system to record routine operational data like machine run-time, worker attendance, and material issuance in real time. Which system is this?
A. Transaction Processing System (TPS). B. Executive Support System (ESS).
C. Office Automation System (OAS). D. Expert System (ES)
 - 7 Which of the following is the best definition of Smart Manufacturing (SM)?
A. Manual production using skilled labor only
B. Intelligent, interconnected, data-driven transformation using advanced digital technologies
C. A system focused only on mass production using assembly lines
D. A method of increasing production by adding more workers
 - 8 ERP–MES integration mainly helps to
A. Stop data sharing. B. Sync business orders with shop-floor production.
C. Replace sensors. D. Increase manual reporting

- 9 What are the three main pillars for IoT in a smart factory?
A. Speed, Cost, Design. B. Security, Privacy, Governance
C. PLC, SCADA, CNC. D. AI, AR, VR
- 10 Which is best to protect data sent from sensors to cloud
A. HTTP only. B. TLS/DTLS encryption.
C. Plain text messaging. D. FTP transfer
- 11 Digital Twin means
A. A real machine copy. B. A virtual copy of a physical asset
C. A wireless sensor. D. A PLC program
- 12 Smart Manufacturing is also called as
A. Industry 1.0. B. Industry 4.0
C. Industry 2.0. D. Industry 3.0
- Q2. Solve the following.
- A) Explain Different IoT Protocols. 6
- B) Explain different Cloud Computing service models. 6
- Q3. Solve the following.
- A) Explain the general architecture of Machine-to-Machine (M2M) system. 6
- B) Explain the difference between Internet of Things (IoT) and Machine-to-Machine (M2M) systems. 6
- Q4. Solve Any Two of the following.
- A) Describe the need for information systems in a manufacturing organization. 6
- B) Explain the different Dimensions of Information Systems in the Manufacturing Industry. 6
- C) Explain in detail the Information Technology Infrastructure in the Manufacturing Industry. 6
- Q5. Solve Any Two of the following.
- A) Explain Smart Manufacturing and its core pillars. 6
- B) Explain the Demand-driven Supply Chain (DDSC) of Smart manufacturing. 6
- C) Explain the Intelligent Energy Demand Management (IEDM) and its key components. 6
- Q6. Solve Any Two of the following.
- A) Explain the Privacy and its issues for Internet of Things (IoT) in Smart Manufacturing. 6
- B) Explain the concept of Security and its issues for Internet of Things (IoT) in Smart Manufacturing. 6
- C) Explain the First Steps Towards a Secure Data Platform for Smart Manufacturing. 6

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